

Eric Ponce

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Experience

E-Circuit Motors, Inc., R&D Director

Jan 2024 – Present | Bozeman, MT

- Lead the design and development of custom BLDC controller hardware (STM32) and firmware, architected as reference designs for rapid customer adoption, with interfacing via a variety of communication protocols (CAN/CANopen, UART, PWM, etc.)
- Take reference designs from concept to production, shipping custom BLDC controllers to customers for real-world deployment
- Drive innovation in advanced motor control strategies, including FOC, sensorless algorithms, parameter estimation, and torque optimization, validated through custom motor, system, and control-algorithm simulations in Python and QSPICE
- Build in-house electronics manufacturing and high-power motor test infrastructure ($\approx 15\text{kW}$ back-to-back setup) to support internal development and customer manufacturing/validation
- Implement demonstration and UI systems showcasing core motor technologies in real-world applications (robotics, haptics, exercise equipment) to accelerate customer testing and engagement
- Prototype hardware and firmware to expand customer sales across all application areas (HVAC, pumps, UUV, robotics, manufacturing, consumer, etc.), and partner directly with keystone clients to diagnose and resolve complex technical issues
- Author white papers, co-author patents, and produce controller documentation and reference manuals to drive customer engagement and adoption

MIT – Electromechanical Systems Group (Prof. Steven Leeb), Graduate Researcher

Jun 2017 – Jan 2024 | Cambridge, MA

- Designed physical models and low-power mixed-signal embedded circuitry (Infineon PSoC), firmware, and software for non-intrusively measuring the state and operation of mechanical systems, such as water meters, to cheaply retrofit for smart resource tracking and diagnostic capability
- Built models, embedded circuitry (Infineon PSoC, Microchip SAML10), and firmware for harvesting ambient energy to enable sensing of systems
- Developed methods for analyzing the stability of grid-connected constant-power-loads which may cause serious instabilities
- Re-architected firmware and built a new web software UI to drastically reduce power consumption and demonstrate new sensing capabilities for research group's wireless sensing platform centered around a TI Tiva C Series microcontroller and CC3100 Wi-Fi network processor
- Mentored and managed undergraduate and graduate researchers on development for research group topics and electrical engineering courses for students across a range of levels (middle-school through university)
- Teaching assistant for four semesters of Power Electronics and Microcomputer Lab. Managed staff hours, laboratory resources, and safety, maintained lab equipment, and helped students learn the material and debug their work (6.96/7.0 rating)

Conduit Tech, Electrical Engineering Consultant

Jun 2022 – Aug 2022 | Cambridge, MA

- Developed hardware/firmware and defined sensor requirements for real-time AWS IoT Core-connected (Wi-Fi/NB-IoT) sensing platform for HVAC

WaterBit, Electrical Engineering Intern

Jun 2020 – Aug 2020 | San Jose, CA

- Developed real-time embedded signal processing methods to non-intrusively detect disastrous valve failure in harsh agricultural environments
- Restructured and rewrote ultra-low power firmware to enable development of new features and reduce technical debt
- Customized the firmware build system for our Infineon PSoC target to improve version control and continuous integration efforts

ORI, Electrical Engineering Consultant

Sep 2016 – Feb 2017 | Cambridge, MA

- Developed highly configurable and easily expandable embedded circuitry for control of movable furniture centered around an Infineon PSoC

Apple – iPad Systems, Electrical Engineering Intern

Jun 2016 – Aug 2016 | Cupertino, CA

- Developed software and tooling and interfaced with vendor to validate, debug, and improve early-stage USB power-delivery
- Validated power distribution and high-speed communication networks and interfaces
- Owned design and production of mixed-signal high-speed multi-layer PCB to enable the testing of high pixel density LED displays
- Provided engineering support in Shenzhen during bring up of large-scale design verification build

Education – Massachusetts Institute of Technology

PhD in Electrical Engineering and Computer Science

GPA: 5.0/5.0

Jun 2019 – Sep 2023

MEng in Electrical Engineering and Computer Science

GPA: 4.8/5.0

Jun 2017 – Jun 2019

BS in Electrical Science and Engineering

GPA: 4.9/5.0

Aug 2013 – Jun 2017

Skills

Hardware	Analog, digital, embedded, low-power, and power electronic circuit design, KiCad, Altium, Cadence, SPICE
Firmware	Communications (SPI, I2C, CAN, Wi-Fi, BT, NB-IoT, etc.), signal processing, multiple MCU architectures (8051, AVR, ARM Cortex-M)
Software	C, Python (NumPy, SciPy, Matplotlib), shell scripting (bash), Git, build system (GNU), HTML, CSS, JavaScript
Other	Numerical & analytical modeling, optimization, 3D printing, laser cutting, solder rework, Spanish

Selected Publications

- Ponce, E., & Leeb, S., "Impedance Modeling for Line-Frequency Rectified Constant-Power Loads," *IEEE Access*, 2025
- Ponce, E., & Leeb, S., "Envelope Impedance Modeling of Line Frequency Rectifiers," *IEEE Trans. on Inds. Elec.*, 2025
- Monagle, D., Ponce, E., & Leeb, S., "Rule the Joule: An Energy Management Design Guide for Self-Powered Sensors," *IEEE Sensors*, 2023
- Ponce, E. & Leeb, S., "Passive Probe: Mechanically-Modulated Field Sensing for Motion Tracking and Flow Estimation," *IEEE Sensors Journal*, 2022
- Ponce, E., Leeb, S., & Lindahl, P., "Know the Flow: Non-Contact Magnetic Flow Rate Sensing for Water Meters," *IEEE Sensors Journal*, 2021
- Ponce, E. & Leeb, S., "A 3-D Field Solution for Axially Polarized Multi-Pole Ring PMs and its Appl. in Position Meas," *IEEE Trans. Magn.*, 2020